

words	description
Algorithm	A set of instructions designed to perform a specific task or solve a problem.
Variable	A named storage location in memory that holds data.
Function	A block of reusable code that performs a specific task.
Loop	A programming construct that repeats a block of code until a certain condition is met.
Conditional	A programming construct that executes different code based on the evaluation of a condition.
Object	An instance of a class in object-oriented programming that encapsulates data and behavior.
Class	A blueprint for creating objects that defines their attributes and methods.
Method	A function associated with an object in object-oriented programming.
Interface	A contract that defines the methods an object must implement.
Inheritance	A mechanism in object-oriented programming where a class can inherit attributes and methods from another class.
Encapsulation	The bundling of data and methods that operate on the data into a single unit (class).
Polymorphism	The ability of objects to be treated as instances of their parent class.
Abstraction	The process of hiding the implementation details and showing only the essential features of an object.
Module	A file containing Python code, typically containing functions, classes, and variables.

Package	A collection of related Python modules that can be easily distributed and reused.
Library	A collection of precompiled routines that a program can use.
API (Application Programming Interface)	A set of rules and protocols that allows different software applications to communicate with each other.
IDE (Integrated Development Environment)	A software application that provides comprehensive facilities to computer programmers for software development.
Debugging	The process of identifying and fixing errors (bugs) in a computer program.
Version Control	The management of changes to documents, computer programs, large websites, and other collections of information.
Repository	A central location in which data is stored and managed.
Git	A distributed version control system used for tracking changes in source code during software development.
Branch	A parallel version of a repository's codebase, allowing changes to be made without affecting the main codebase.
Merge	The process of combining changes from different branches into one branch.
Commit	A record of changes to a repository, saved with a unique identifier.
Pull Request	A method of submitting contributions to a repository in distributed version control systems.
Dependency	A file or package required for a software project to function correctly.
Framework	A collection of pre-written code that provides a structure for developers to build upon.
Compiler	A program that translates source code written in a high-level programming language into machine code.

Interpreter	A program that directly executes instructions written in a programming or scripting language.
Bytecode	Intermediate code generated by compilers and executed by virtual machines.
API Gateway	A server that acts as an API front-end, receiving API requests, enforcing throttling and security policies, passing requests to the back-end service, and then passing the response back to the requester.
CRUD Operations	Create, Read, Update, Delete - the four basic functions of persistent storage.
SQL (Structured Query Language)	A domain-specific language used in programming and designed for managing data held in a relational database management system (RDBMS).
NoSQL	A type of database that provides a mechanism for storage and retrieval of data that is modeled in means other than the tabular relations used in relational databases.
RESTful	Representational State Transfer, a software architectural style that defines a set of constraints to be used for creating web services.
JSON (JavaScript Object Notation)	A lightweight data-interchange format that is easy for humans to read and write and easy for machines to parse and generate.
XML (eXtensible Markup Language)	A markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable.
Scalability	The capability of a system, network, or process to handle a growing amount of work, or its potential to be enlarged to accommodate that growth.
Agile	A software development methodology based on iterative and incremental development, where requirements and solutions evolve through collaboration between self-organizing cross-functional teams.
Waterfall	A traditional sequential software development model in which progress is seen as flowing steadily downwards through the phases of conception, initiation, analysis, design, construction, testing, production/implementation, and maintenance.
DevOps	A set of practices that combines software development (Dev) and information-technology operations (Ops) that aims to shorten the systems development life cycle and provide continuous delivery with high software quality.
Continuous Integration (CI)	The practice of automatically testing and integrating code changes into a shared repository frequently.
Continuous Deployment (CD)	The practice of automatically deploying every code change to production environments after passing through the automated tests.

Docker	A platform for developing, shipping, and running applications inside containers.
Kubernetes	An open-source container orchestration platform for automating the deployment, scaling, and management of containerized applications.
Microservices	An architectural style that structures an application as a collection of loosely coupled services.
Scalability	The ability of a system, network, or process to handle a growing amount of work.
Authentication	The process of verifying the identity of a user or system.
Authorization	The process of determining whether a user or system has permission to access certain resources or perform certain actions.
Abstraction	The process of hiding complex implementation details and exposing only the essential features of an object or system.
Algorithm	A set of step-by-step instructions designed to solve a specific problem or perform a specific task.
Analysis	The process of studying and evaluating the efficiency, performance, and complexity of algorithms.
Big O Notation	A mathematical notation used to describe the limiting behavior of a function when the argument tends towards a particular value or infinity, often used to analyze the time complexity of algorithms.
Binary Search	A search algorithm that finds the position of a target value within a sorted array by repeatedly dividing the search interval in half.
Branching	A control flow statement that allows a program to execute different sequences of code based on specified conditions.
Bubble Sort	A simple sorting algorithm that repeatedly steps through the list, compares adjacent elements, and swaps them if they are in the wrong order.
Data Structure	A way of organizing and storing data in a computer so that it can be accessed and modified efficiently.
Dijkstra's Algorithm	An algorithm for finding the shortest paths between nodes in a graph, which may represent, for example, road networks.

Dynamic Programming	A method for solving complex problems by breaking them down into simpler subproblems and solving each subproblem only once.
Efficiency	The ability to achieve maximum productivity with minimum wasted effort or expense, often measured in terms of time or space complexity in algorithms.
Encapsulation	The bundling of data and methods that operate on the data into a single unit (class) in object-oriented programming.
Function	A self-contained block of code that performs a specific task and can be reused throughout a program.
Graph	A collection of nodes (vertices) and edges that connect pairs of nodes, representing relationships between objects.
Hash Table	A data structure that stores key-value pairs and provides fast insertion, deletion, and lookup operations.
Heap Sort	A comparison-based sorting algorithm that creates a binary heap from the input data and then repeatedly extracts the maximum element from the heap and rebuilds the heap until the heap is empty.
Insertion Sort	A simple sorting algorithm that builds the final sorted array one element at a time by repeatedly inserting the next element into the already sorted part of the array.
Iteration	The process of repeating a set of instructions a specified number of times or until a specific condition is met.
Linked List	A data structure consisting of a sequence of elements (nodes), where each node contains a reference to the next node in the sequence.
Merge Sort	A divide-and-conquer sorting algorithm that divides the input array into two halves, sorts each half recursively, and then merges the sorted halves.
Object-Oriented Programming (OOP)	A programming paradigm based on the concept of "objects", which can contain data in the form of fields (attributes) and code in the form of procedures (methods).
Polymorphism	The ability of objects to take on different forms or behaviors based on their context or the data they operate on.
Queue	A data structure that follows the First-In-First-Out (FIFO) principle, where elements are inserted at the rear and removed from the front.
Recursion	The process of a function calling itself directly or indirectly in order to solve a problem that can be broken down into smaller, similar subproblems.

Selection Sort	A simple sorting algorithm that divides the input array into two parts: the sorted part and the unsorted part, and repeatedly selects the smallest (or largest) element from the unsorted part and moves it to the beginning of the sorted part.
Searching	The process of finding a specific item (key) in a collection of items (data structure) such as an array, list, or tree.
Sorting	The process of arranging items in a specific order, often numerical or lexicographical order, based on a key value.
Stack	A data structure that follows the Last-In-First-Out (LIFO) principle, where elements are inserted and removed from the same end, typically called the top.
Tree	A hierarchical data structure consisting of nodes connected by edges, where each node has a parent node and zero or more child nodes.
Recursion	The process of a function calling itself directly or indirectly in order to solve a problem that can be broken down into smaller, similar subproblems.
Quick Sort	A divide-and-conquer sorting algorithm that selects a pivot element from the array and partitions the array into two sub-arrays, such that elements less than the pivot are on the left and elements greater than the pivot are on the right. It then recursively sorts the sub-arrays.
Binary Tree	A tree data structure in which each node has at most two children, referred to as the left child and the right child.
Binary Search Tree (BST)	A binary tree data structure in which each node has a key value, and the key values in the left subtree are less than the key value of the root, and the key values in the right subtree are greater than the key value of the root.
Graph Traversal	The process of visiting all the nodes of a graph in a systematic way, often used to search for specific elements or paths within the graph.
Greedy Algorithm	An algorithmic paradigm that follows the problem-solving heuristic of making the locally optimal choice at each stage with the hope of finding a global optimum solution.
Hashing	The process of mapping data of arbitrary size to data of a fixed size, typically used to index and retrieve items in a data structure such as a hash table.
In-place Algorithm	An algorithm that does not require extra space and operates directly on the input data, modifying it in place.
NP-Complete Problem	A class of problems for which no efficient algorithm exists that can solve all instances of the problem in polynomial time.
NP-Hard Problem	A class of problems for which no efficient algorithm exists that can solve all instances of the problem in polynomial time, but for which a solution can be verified in polynomial time.

Optimization	The process of making a system or design as effective or functional as possible, often involving trade-offs between different factors such as time, space, and complexity.
Parallelism	The use of multiple processing units to perform computations simultaneously, improving performance and efficiency.
Pseudocode	A high-level description of a computer program or algorithm written in plain language, often used for planning and communicating ideas before actual coding begins.
Queue	A data structure that follows the First-In-First-Out (FIFO) principle, where elements are inserted at the rear and removed from the front.
Radix Sort	A non-comparative sorting algorithm that sorts integers by grouping them by individual digits that share the same significant position and value.
Red-Black Tree	A type of self-balancing binary search tree that maintains balance during insertions and deletions by enforcing certain properties.
Run-Time Complexity	The amount of time an algorithm takes to complete as a function of the size of the input data.
Space Complexity	The amount of memory an algorithm requires to execute as a function of the size of the input data.
Suffix Array	A sorted array of all suffixes of a given string, often used in various string manipulation and search algorithms.
Trie	A data structure used to store a dynamic set of strings, where each node represents a common prefix of strings.
Two-Pointer Technique	A programming technique that uses two pointers to solve problems involving arrays or linked lists, often used to find pairs of elements that satisfy certain conditions.
Algorithm	A set of instructions designed to perform a specific task or solve a problem.
API (Application Programming Interface)	A set of rules and protocols that allows different software applications to communicate with each other.
Backend	The part of a web application or software system that runs on the server and interacts with the database and other resources.
Bug	An error, flaw, or fault in a computer program or system that causes it to produce an incorrect or unexpected result.

Cache	A hardware or software component that stores data temporarily to reduce the time it takes to access data from a primary storage location.
Client-side	Refers to the part of a web application or software system that runs on the client's computer or device, typically in a web browser.
CSS (Cascading Style Sheets)	A style sheet language used for describing the presentation of a document written in HTML.
Debugging	The process of identifying and fixing errors (bugs) in a computer program or system.
Deployment	The process of making a software application or system available for use by deploying it to a server or hosting environment.
Framework	A collection of pre-written code that provides a structure for developers to build upon, speeding up the development process.
Frontend	The part of a web application or software system that users interact with directly in their web browsers.
Git	A distributed version control system used for tracking changes in source code during software development.
IDE (Integrated Development Environment)	A software application that provides comprehensive facilities to computer programmers for software development.
Library	A collection of precompiled routines that a program can use, providing specific functionality.
Metadata	Data that describes other data, providing information about the content, quality, condition, origin, and other characteristics of data.
Module	A file containing code or definitions that can be imported and used in another Python program.
Package	A collection of related modules that can be easily distributed and reused in Python.
Patch	A small update to a software application or system that fixes a specific issue or problem.
Plugin	A software component that adds specific features or functionality to an existing software application or system.

Repository	A central location in which data is stored and managed, typically used for version control in software development.
Responsive Design	A design approach that ensures a web application's layout and content adapt to the size of the screen or device.
Scalability	The capability of a system, network, or process to handle a growing amount of work.
SEO (Search Engine Optimization)	The process of optimizing a website to improve its visibility and ranking in search engine results.
Syntax	The set of rules that defines the combinations of symbols that are considered to be correctly structured programs or expressions in a programming language.
Template	A preformatted document or file used as a starting point for a specific purpose, such as a website or software application.
UI (User Interface)	The part of a software application or system that users interact with to control and manipulate data and perform tasks.
UX (User Experience)	The overall experience of a person using a product such as a website or software application, encompassing aspects of design, usability, and accessibility.
Version Control	The management of changes to documents, computer programs, large websites, and other collections of information.
Widget	A small, reusable software component or graphical control element used to display information or interact with a user in a graphical user interface.
Wrapper	A software component or function that encapsulates or wraps another component or function to simplify its use or extend its functionality.
Agile	A software development methodology based on iterative and incremental development, where requirements and solutions evolve through collaboration between self-organizing cross-functional teams.
Backlog	A prioritized list of tasks or features to be completed in a software development project.
Continuous Integration	The practice of automatically testing and integrating code changes into a shared repository frequently.
Cross-platform	Refers to software or applications that can run on multiple operating systems or platforms.

Dead Code	Code that is no longer used or executed in a software application but remains in the codebase, potentially causing maintenance issues or performance problems.
Endpoint	A specific URL or URI that represents a resource in a RESTful API.
Inheritance	A mechanism in object-oriented programming where a class can inherit attributes and methods from another class.
Lambda Function	An anonymous function in Python defined using the lambda keyword.
Microservices	An architectural style that structures an application as a collection of loosely coupled services.
Refactoring	The process of restructuring existing code without changing its external behavior to improve readability, maintainability, or performance.
Regression Testing	The process of retesting a software application or system to ensure that recent code changes have not adversely affected existing features or functionality.
Sprint	A short, time-boxed period in which a team works to complete a set amount of work or achieve a specific goal in a software development project.
Stub	A placeholder or simulated implementation used in testing to replace a module or function that has not yet been implemented.
Unit Testing	The process of testing individual units or components of a software application or system to ensure that they function correctly in isolation.
Wrapper	A software component or function that encapsulates or wraps another component or function to simplify its use or extend its functionality.
YAML (YAML Ain't Markup Language)	A human-readable data serialization standard used for configuration files and data exchange in software applications.
Singleton	A design pattern that restricts the instantiation of a class to a single object, ensuring that there is only one instance of the class in the system.
Proxy Server	A server that acts as an intermediary between clients and other servers, intercepting requests and forwarding them to the appropriate destination.
Mocking	The process of creating simulated objects or components to replace real objects or components during testing.

Scrum	A framework for agile software development that emphasizes teamwork, accountability, and iterative progress towards a common goal.
HTML	(Hypertext Markup Language) - A markup language used for creating and structuring web pages and web applications.
CSS	(Cascading Style Sheets) - A style sheet language used for describing the presentation of a document written in HTML.
JavaScript	A programming language used for adding interactivity and dynamic behavior to web pages.
DOM	(Document Object Model) - A programming interface for HTML and XML documents, representing the structure of the document as a tree of objects.
API	(Application Programming Interface) - A set of rules and protocols that allows different software applications to communicate with each other.
HTTP	(Hypertext Transfer Protocol) - The protocol used for transmitting data over the World Wide Web.
HTTPS	(Hypertext Transfer Protocol Secure) - An extension of HTTP that uses encryption to secure the communication between the client and the server.
URL	(Uniform Resource Locator) - A reference to a web resource that specifies its location and how to access it.
URI	(Uniform Resource Identifier) - A string of characters used to identify a resource on the internet.
DNS	(Domain Name System) - A hierarchical and decentralized naming system for computers, services, or other resources connected to the internet.
Web Server	A software application or hardware device that serves content to clients over the internet.
Client	A computer or device that requests data or services from a server.
Server	A computer or device that provides data or services to clients over a network.
Frontend	The part of a web application that users interact with directly in their web browsers.

Backend	The part of a web application that runs on the server and interacts with the database and other resources.
MVC	(Model-View-Controller) - A software architectural pattern for implementing user interfaces, dividing an application into three interconnected components.
REST	(Representational State Transfer) - An architectural style for designing networked applications, typically using HTTP requests to perform CRUD operations on resources.
CRUD	(Create, Read, Update, Delete) - The four basic functions of persistent storage in web applications.
AJAX	(Asynchronous JavaScript and XML) - A technique for creating interactive web applications by asynchronously sending and receiving data from the server.
JSON	(JavaScript Object Notation) - A lightweight data-interchange format that is easy for humans to read and write and easy for machines to parse and generate.
XML	(eXtensible Markup Language) - A markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable.
Web Application	A software application that runs on a web server and is accessed through a web browser over the internet.
Web Browser	A software application used to access and view websites on the World Wide Web.
Responsive Design	A design approach that ensures a web application's layout and content adapt to the size of the screen or device.
Mobile-First Design	A design approach that prioritizes designing and developing for mobile devices before desktop computers.
Framework	A collection of pre-written code that provides a structure for developers to build upon, speeding up the development process.
Library	A collection of precompiled routines that a program can use, providing specific functionality.
CMS	(Content Management System) - A software application or set of related programs that are used to create and manage digital content.
SEO	(Search Engine Optimization) - The process of optimizing a website to improve its visibility and ranking in search engine results.

SEM	(Search Engine Marketing) - A form of internet marketing that involves promoting websites by increasing their visibility in search engine results pages.
Analytics	The collection, measurement, analysis, and reporting of web data to understand and optimize web usage.
Cookies	Small pieces of data stored on the client's computer by the web browser, used to track user activity and personalize content.
Session	A period of time during which a user interacts with a web application, typically starting when the user logs in and ending when the user logs out.
Authentication	The process of verifying the identity of a user or system.
Authorization	The process of determining whether a user or system has permission to access certain resources or perform certain actions.
Scalability	The capability of a system, network, or process to handle a growing amount of work.
Accessibility	The practice of ensuring that web content is accessible to all users, including those with disabilities.
Usability	The ease of use and learnability of a web application's interface.
Wireframe	A visual representation of the layout and structure of a web page or application.
Prototype	A preliminary version of a web application used to test and evaluate its design and functionality.
Version Control	The management of changes to documents, computer programs, large websites, and other collections of information.
Git	A distributed version control system used for tracking changes in source code during software development.
Deployment	The process of making a web application available for use by deploying it to a server or hosting environment.
Continuous Integration	The practice of automatically testing and integrating code changes into a shared repository frequently.

Continuous Deployment	The practice of automatically deploying every code change to production environments after passing through automated tests.
DevOps	A set of practices that combines software development (Dev) and information-technology operations (Ops) to shorten the systems development life cycle.
Web Socket	A communication protocol that provides full-duplex communication channels over a single TCP connection.
CDN	(Content Delivery Network) - A geographically distributed network of servers that delivers web content to users based on their location.
SSL	(Secure Sockets Layer) - A standard security protocol for establishing encrypted links between a web server and a browser in an online communication.
TLS	(Transport Layer Security) - The successor to SSL, a cryptographic protocol that provides secure communication over a computer network.
CPython	The default and most widely used implementation of the Python programming language, written in C.
Cython	A superset of Python that allows calling C functions and declaring C types, enabling high-performance Python code.
GIL	(Global Interpreter Lock) - A mutex that protects access to Python objects, preventing multiple native threads from executing Python bytecodes simultaneously in CPython .
ctypes	A foreign function library for Python that provides C compatible data types and allows calling functions in DLLs or shared libraries.
SWIG	(Simplified Wrapper and Interface Generator) - A software development tool that connects programs written in C or C++ with scripting languages like Python.
FFI	(Foreign Function Interface) - A mechanism that allows code written in one programming language to call functions or access data written in another language.
PyPy	An alternative implementation of the Python programming language with a focus on speed and memory efficiency, written in Python and translated to machine code using a JIT compiler.
JIT	(Just-In-Time) Compiler - A compiler that converts bytecode into machine code at runtime, improving the performance of interpreted languages like Python.
AST	(Abstract Syntax Tree) - A hierarchical representation of the syntactic structure of code written in a programming language, used for analysis and transformation.

Bytecode	The intermediate representation of Python code compiled by the Python interpreter before execution.
Disassembly	The process of converting bytecode back into human-readable Python code.
Stack Frame	A data structure that stores information about the execution state of a function, including local variables, parameters, and return addresses.
Frame Object	A Python object representing the execution state of a function, used by the Python interpreter to manage function calls and returns.
C API	The Application Programming Interface provided by CPython for extending Python with C code.
Extension Module	A Python module written in C or C++ that extends the functionality of Python by providing low-level access to system calls and libraries.
Memory Management	The process of allocating and deallocating memory for objects and data structures in a computer program.
Reference Counting	A memory management technique used by CPython to track the number of references to an object and automatically deallocate memory when the reference count drops to zero.
Garbage Collection	The process of reclaiming memory occupied by objects that are no longer in use, performed by the Python interpreter's garbage collector.
Weak Reference	A reference to an object that does not prevent the object from being garbage collected if there are no strong references to it.
C Extension	A Python module written in C or C++ that extends the functionality of Python by providing access to low-level system calls and libraries.
GIL	(Global Interpreter Lock) - A mutex that protects access to Python objects, preventing multiple native threads from executing Python bytecodes simultaneously in CPython .
Multi-Threading	The ability of a program to execute multiple threads concurrently within the same process.
Multi-Processing	The ability of a program to execute multiple processes concurrently, allowing for true parallelism on multi-core systems.
Concurrency	The ability of a program to execute multiple tasks simultaneously, regardless of whether they run on separate threads or processes.

Asynchronous I/O	A programming paradigm that allows I/O operations to be performed concurrently with other tasks, improving the responsiveness and scalability of applications.
Coroutine	A generalization of subroutines for non-preemptive multitasking, allowing multiple entry points for suspending and resuming execution at specific locations.
Context Switching	The process of saving and restoring the state of a CPU so that multiple processes or threads can share a single CPU resource.
Race Condition	A situation that occurs when the outcome of a program depends on the relative timing or interleaving of operations performed by multiple threads or processes.
Locking	The process of synchronizing access to shared resources using locks to prevent race conditions and ensure data integrity.
Deadlock	A situation that occurs when two or more processes are unable to proceed because each is waiting for the other to release a resource.
Semaphore	A synchronization primitive used to control access to a shared resource by multiple processes or threads.
Thread Pool	A collection of pre-initialized threads that are ready to perform tasks, improving the efficiency of thread creation and management.
Futures	A high-level abstraction for asynchronous programming, representing the result of a computation that may not yet be complete.
Event Loop	A programming construct that continuously monitors and dispatches events or messages in an application, typically used in asynchronous I/O frameworks.
Callback	A function that is passed as an argument to another function and is executed after a specific event occurs.
Event-Driven Programming	A programming paradigm that focuses on handling events or messages generated by user actions or system events.
Coroutine	A special type of function that can suspend and resume execution at specific points, allowing for cooperative multitasking and asynchronous programming.
Futures	A high-level abstraction for asynchronous programming, representing the result of a computation that may not yet be complete.
Promise	A proxy for a value that may not yet be available, allowing asynchronous operations to be composed and chained together.

Asncio	A library in Python that provides support for writing asynchronous I/O code using the <code>async/await</code> syntax.
Concurrent.futures	A module in Python that provides a high-level interface for asynchronously executing callables in a <code>ThreadPoolExecutor</code> or <code>ProcessPoolExecutor</code> .
Threading	A module in Python that provides support for creating and managing threads for concurrent execution.
Multiprocessing	A module in Python that provides support for creating and managing processes for parallel execution.
Async/await	A syntax introduced in Python 3.5 for defining asynchronous functions and awaiting the results of asynchronous operations.
Coroutine	A special type of function that can suspend and resume execution at specific points, allowing for cooperative multitasking and asynchronous programming.
Asynchronous Generator	A special type of generator that produces values asynchronously using the <code>async/await</code> syntax.
Event Loop	A programming construct that continuously monitors and dispatches events or messages in an application, typically used in asynchronous I/O frameworks.
Callback	A function that is passed as an argument to another function and is executed after a specific event occurs.
Promise	A proxy for a value that may not yet be available, allowing asynchronous operations to be composed and chained together.
Concurrency	The ability of a program to execute multiple tasks simultaneously, regardless of whether they run on separate threads or processes.
Database	A structured collection of data that is organized and stored for easy access, retrieval, and management.
SQL (Structured Query Language)	A domain-specific language used in programming and designed for managing data held in a relational database management system (RDBMS).
Table	A collection of related data organized into rows and columns in a database.
Row	A single record in a database table that contains data about a specific entity.

Column	A vertical arrangement of data in a database table representing a specific attribute of the entity.
Primary Key	A unique identifier for each record in a database table, used to uniquely identify each row.
Foreign Key	A column or set of columns in a database table that refers to the primary key of another table, establishing a relationship between the two tables.
Index	A data structure in a database that improves the speed of data retrieval operations on a table.
Query	A request for data or information from a database, typically written in SQL.
Relational Database	A type of database that stores and organizes data into tables and establishes relationships between them.
NoSQL Database	A type of database that provides a mechanism for storage and retrieval of data that is modeled in means other than the tabular relations used in relational databases.
Database Management System (DBMS)	Software that allows users to interact with a database, providing functionalities for data storage, retrieval, and manipulation.
ACID (Atomicity, Consistency, Isolation, Durability)	A set of properties that guarantee reliable transactions in a database system.
Transaction	A single unit of work performed within a database management system.
Normalization	The process of organizing the attributes and tables of a relational database to minimize redundancy and dependency.
Denormalization	The process of adding redundant data to a normalized database to improve performance.
Backup	A copy of data kept separate from the original data to protect against loss or damage.
Restore	The process of recovering data from a backup and returning it to its original state.
Replication	The process of copying and distributing data from one database to another for redundancy, backup, or load balancing purposes.

Sharding	A database partitioning technique that divides a large database into smaller, more manageable parts called shards.
Partitioning	The process of dividing a large database table into smaller, more manageable parts.
Data Warehouse	A centralized repository for storing and analyzing large volumes of structured and unstructured data.
Data Mart	A subset of a data warehouse that is designed for a specific business line or department.
OLTP (Online Transaction Processing)	A class of systems that facilitate and manage transaction-oriented applications, typically used for data entry and retrieval in real-time.
OLAP (Online Analytical Processing)	A class of systems that facilitate and manage complex queries and analysis of data stored in a data warehouse.
ETL (Extract, Transform, Load)	A process used to extract data from various sources, transform it into a format suitable for analysis, and load it into a data warehouse.
Data Mining	The process of discovering patterns and insights from large datasets using statistical, mathematical, and machine learning techniques.
Big Data	A term used to describe large volumes of data, both structured and unstructured, that cannot be easily processed using traditional database and software techniques.
Hadoop	An open-source framework for distributed storage and processing of large datasets across clusters of computers.
MapReduce	A programming model and processing technique for parallel processing of large datasets across distributed clusters.
Spark	An open-source cluster computing framework that provides an interface for programming entire clusters with implicit data parallelism and fault tolerance.
Columnar Database	A database management system that stores data by column rather than by row, allowing for faster queries and better compression.
Document Database	A type of NoSQL database that stores and retrieves data in a document-oriented format, such as JSON or XML.
Key-Value Store	A type of NoSQL database that stores data as a collection of key-value pairs.

Graph Database	A type of database that uses graph structures for semantic queries with nodes, edges, and properties to represent and store data.
CAP Theorem (Consistency, Availability, Partition Tolerance)	A theorem that states that it is impossible for a distributed data store to simultaneously provide more than two out of three guarantees: consistency, availability, and partition tolerance.
Data Model	A conceptual representation of data objects, relationships between data, and rules governing those relationships.
Concurrency Control	The management of simultaneous access to shared resources in a database to prevent inconsistencies.
Data Dictionary	A centralized repository of metadata, containing descriptions and definitions of data elements, data structures, and data flows.
Data Governance	The overall management of the availability, usability, integrity, and security of data used in an enterprise.
Master Data Management (MDM)	A process that ensures uniformity, accuracy, stewardship, and accountability of enterprise master data assets.
Backup and Recovery	The processes and strategies for protecting data against loss and restoring it in the event of a failure or disaster.
Data Encryption	The process of converting data into a form that cannot be easily understood by unauthorized users.
Data Compression	The process of reducing the size of data to save storage space and improve data transfer speeds.
Data Warehousing	The process of collecting, storing, and managing large volumes of structured and unstructured data from various sources for analysis and reporting.
Data Cleansing	The process of identifying and correcting errors and inconsistencies in data to improve its quality and reliability.
Data Lake	A centralized repository that allows you to store all your structured and unstructured data at any scale, securely and cost-effectively.
Data Mining	The process of discovering patterns and insights from large datasets using statistical, mathematical, and machine learning techniques.
Data Visualization	The graphical representation of information and data to facilitate understanding and decision-making.

Data Integration	The process of combining data from different sources into a unified view to provide a comprehensive and accurate representation of information.
Python	A high-level, interpreted programming language known for its simplicity and readability.
Interpreter	A program that translates and executes source code written in a high-level programming language like Python.
Compiler	A program that translates source code written in a high-level programming language into machine code.
PEP (Python Enhancement Proposal)	A design document providing information to the Python community or describing a new feature for Python or its processes or environment.
Syntax	The set of rules that defines the combinations of symbols that are considered to be correctly structured programs or expressions in Python.
Indentation	The whitespace at the beginning of a line of code that indicates the block structure in Python.
Dynamic Typing	A programming language feature in which the type of a variable is determined at runtime, as opposed to compile time.
Static Typing	A programming language feature in which the type of a variable is explicitly declared and checked at compile time.
Object-Oriented Programming (OOP)	A programming paradigm based on the concept of "objects", which can contain data in the form of fields (attributes) and code in the form of procedures (methods).
Class	A blueprint for creating objects in Python, which defines attributes and methods.
Object	An instance of a class in Python that encapsulates data and behavior.
Inheritance	A mechanism in object-oriented programming where a class can inherit attributes and methods from another class.
Method	A function associated with an object in Python.
Function	A block of reusable code in Python that performs a specific task.

Module	A file containing Python code, typically containing functions, classes, and variables.
Package	A collection of related Python modules that can be easily distributed and reused.
Library	A collection of prewritten code in Python that provides specific functionality.
Namespace	A container for a set of identifiers (such as variable names) in Python.
Global	A variable or identifier that is accessible throughout the entire Python program.
Local	A variable or identifier that is accessible only within a specific scope, such as a function.
List	A mutable, ordered collection of items in Python, represented by square brackets [].
Tuple	An immutable, ordered collection of items in Python, represented by parentheses () .
Dictionary	A mutable, unordered collection of key-value pairs in Python, represented by curly braces {}.
Set	An unordered collection of unique elements in Python, represented by curly braces {}.
Slice	A mechanism in Python for extracting a subset of elements from a sequence like a list or tuple.
Indexing	The process of accessing individual elements within a sequence like a list or tuple in Python.
Iteration	The process of looping over elements in a collection, such as a list, tuple, or dictionary, in Python.
Comprehension	A concise way to create lists, dictionaries, and sets in Python using a single expression.
Lambda	An anonymous function in Python defined using the lambda keyword.

Decorator	A design pattern in Python that allows behavior to be added to functions or classes.
Generator	A function in Python that allows you to iterate over a potentially infinite sequence of items.
Iterator	An object in Python that represents a stream of data, allowing you to iterate over its elements one at a time.
JSON (JavaScript Object Notation)	A lightweight data-interchange format that is easy for humans to read and write and easy for machines to parse and generate in Python.
CSV (Comma-Separated Values)	A file format used to store tabular data, commonly used in Python for data manipulation and analysis.
Exception	An error that occurs during the execution of a Python program, which disrupts the normal flow of the program's instructions.
Try-except	A control flow statement in Python used to handle exceptions.
Virtual Environment	An isolated Python environment that allows you to install packages independently of the system-wide Python installation.
Pip	The package installer for Python, used to install and manage Python packages.
Anaconda	A distribution of Python and R programming languages for scientific computing, used for data science and machine learning projects.
NumPy	A Python library used for numerical computing, providing support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays.
Pandas	A Python library used for data manipulation and analysis, providing data structures and functions to work with structured data.
Matplotlib	A Python library used for creating static, animated, and interactive visualizations in Python.
Scikit-learn	A Python library used for machine learning and data mining, providing tools for classification, regression, clustering, dimensionality reduction, and more.
TensorFlow	An open-source machine learning framework developed by Google, used for building and training deep learning models.

PyTorch	An open-source machine learning library developed by Facebook, used for building and training deep learning models.
Flask	A lightweight web application framework for Python, used to build web applications quickly and easily.
Django	A high-level Python web framework that encourages rapid development and clean, pragmatic design.
WSGI (Web Server Gateway Interface)	A specification for web servers and Python web application frameworks to communicate with web servers.
ORM (Object-Relational Mapping)	A programming technique used to convert data between incompatible type systems using object-oriented programming languages.
RESTful API	An architectural style for designing networked applications, typically using HTTP requests to perform CRUD (Create, Read, Update, Delete) operations on resources.